

Six linear factors and a positive parameter bound: a proposed extension to $k = 3$ in Erdős problem 727

AI-assisted mathematical continuation

17 September 2026

Abstract

The supplied manuscript treats only $k = 2$. This separate continuation uses $210 = 14 \cdot 15$ and $420 = 20 \cdot 21$ to factor all three consecutive integers $n + 1, n + 2, n + 3$ for the family $n = 210t^2 + 391t + 179$. Six simple root classes replace the earlier four. A signed residue calculation at modulus 210 gives an average large-prime obstruction coefficient $5/6$. Combining this with uniform carry estimates gives the proposed lower limiting success proportion $0.255039321662196649\dots$ for $((n + 3)!)^2 \mid (2n)!$. The argument is written below with its changed hypotheses and costs, not inferred from finite scans.

Status and scope. This is a new proposed $k = 3$ argument developed during this review, not a result asserted in the supplied September 16 PDF or the older repository. It has not received independent peer review or formal verification. No claim of priority, accepted resolution, or a result for every fixed k is made. Numerical checks in Section 7 verify finite arithmetic only. All constants, including modulus 210, are fixed before the limiting parameter tends to infinity.

1 The explicit family and proposed bound

Put

$$H(t) = 210t^2 + 391t + 179, \quad m = 210t + 195, \quad n = H(t) = \frac{m(m+1)}{210} - 3. \quad (1)$$

As in the $k = 2$ companion, write

$$S = 4 \sum_{J=4}^{\infty} 2^{1-J} \log \frac{J+1}{J}. \quad (2)$$

All logarithms are natural. Define

$$\rho_3 = 1 - \frac{3}{2}S - \frac{5}{12} \log 3 = 0.2550393216621966492464063134168790975\dots \quad (3)$$

Proposed theorem 1. *The explicit family (1) satisfies*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \# \{t \in [T, 2T] \cap \mathbb{Z} : ((H(t) + 3)!)^2 \mid (2H(t))!\} \geq \rho_3 > 0. \quad (4)$$

Consequently the stated argument, if correct, yields infinitely many distinct positive solutions for $k = 3$.

This statement does not upgrade the parameter bound for the original polynomial $6t^2 + 11t + 3$, nor does it assert any finite interval must attain ρ_3 .

2 All three denominator factors are controlled

The choice of 210 permits the simultaneous identities

$$\begin{aligned} n+1 &= \frac{(m-20)(m+21)}{210} = (6t+5)(35t+36), \\ n+2 &= \frac{(m-14)(m+15)}{210} = (t+1)(210t+181), \\ n+3 &= \frac{m(m+1)}{210} = (14t+13)(15t+14). \end{aligned} \tag{5}$$

Each of the six displayed linear polynomials is primitive. Thus the denominator product $E(n) = (n+1)(n+2)(n+3)$ has exactly these six primitive factors, including multiplicities. Cancelling $(n!)^2$ gives

$$((n+3)!)^2 \mid (2n)! \iff E(n)^2 \mid \binom{2n}{n}. \tag{6}$$

For a prime p , Legendre's formula gives

$$C_p(n) := v_p \binom{2n}{n} = \sum_{j \geq 1} \mathbf{1}_{\{n/p^j \geq 1/2\}}. \tag{7}$$

It suffices to check $C_p(n) \geq 2v_p(E(n))$ at primes dividing $E(n)$.

Let

$$\begin{aligned} \mathcal{C} &= \{0, 1, -14, 15, -20, 21\}, & d_c &= 1 - 2c, \\ i_c &= \begin{cases} 3, & c = 0, 1, \\ 2, & c = -14, 15, \\ 1, & c = -20, 21. \end{cases} \end{aligned}$$

The maximum difference between two shifts is 41. For $p > 41$, the prime is a unit modulo 210 and divides at most one $m+c$. If $p^e \mid m+c$, then $v_p(E(n)) = e$ and $n \equiv -i_c \pmod{p^e}$. Since $i_c \leq 3$ and $p > 41$,

$$\{n/p^j\} = 1 - i_c/p^j > 1/2 \quad (1 \leq j \leq e).$$

There are at least e initial carries. In particular, when $v_p(E(n)) = 1$, failure is equivalent to $C_p(n) = 1$: every later carry must be absent.

For $m+c = ap$, direct substitution gives

$$n = \frac{a^2 p^2 + d_c a p}{210} - i_c, \quad \frac{n}{p^2} = \frac{a^2}{210} + \frac{d_c a}{210p} - \frac{i_c}{p^2}. \tag{8}$$

The six derivative constants are 1, -1, 29, -29, 41, -41, all nonzero modulo primes $p > 41$.

3 Fixed primes, repeated primes and small growing primes

Work on

$$\mathcal{M}_X = \{m \in [X, 2X] \cap \mathbb{Z} : m \equiv 195 \pmod{210}\}, \quad N_X = X/210 + O(1).$$

3.1 Every fixed prime has a density-zero failure set

For each fixed p , primitivity in (5) implies

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \#\{t \in [T, 2T] : v_p(E(H(t))) \geq 6h\} \leq 6p^{-h}.$$

Each factor has either no root or one root modulo p^h . A product valuation at least $6h$ requires a factor valuation at least h .

For fixed K , the number of residues modulo p^L with at most K carries in the first L levels is

$$O_{p,K}(L^K \lceil p/2 \rceil^L) = o(p^L).$$

Indeed a digit at least $\lceil p/2 \rceil$ forces a carry whatever the incoming carry; choose at most K such digit positions. For $p \mid 210$, $H(t) \equiv 391t + 179 \pmod{p}$ is a permutation and $H'(t) = 420t + 391$ is always a unit. Simple lifting makes H a permutation modulo every p^L .

For $p \nmid 210$, the prime is odd and the exact identity

$$840H(t) + 2521 = (420t + 391)^2 \tag{9}$$

reduces the issue to square-root fibers. Discard $p^h \mid 420t + 391$, a class of density p^{-h} . For $L > 2h$ the remaining fibers have at most $2p^h$ elements: a root with valuation $s < h$ reduces to a unit square-root congruence modulo p^{L-2s} , with at most two roots and p^s lifts. Send $L \rightarrow \infty$ first, then $h \rightarrow \infty$. This proves $C_p(H(t)) \rightarrow \infty$ in probability. Combining it with the denominator valuation bound proves the fixed-prime assertion. All primes at most 41 are included here; no simplicity assertion is needed for those primes.

3.2 Repeated nonfixed factors

Fix $P > 41$. For $p > P$, $p^2 \mid m + c$ specifies one class modulo $210p^2$ and requires $p \leq \sqrt{2X + 21}$. Summing the progression count $X/(210p^2) + O(1)$ over six shifts and these primes gives

$$O(X/P) + O(\sqrt{X}).$$

Its normalized limsup is $O(1/P)$. Outside this exception and the fixed-prime exceptions, all obstructing primes have $v_p(E(n)) = 1$.

3.3 Small growing primes

For $P < p \leq X^\delta$, $0 < \delta < 1/4$, the six root classes in the t variable are distinct and simple, with derivatives d_c modulo p . On one root class, $H(t)$ maps bijectively modulo p^L onto $n \equiv -i_c \pmod{p}$. Take $L = \lfloor \log_p(X/420) \rfloor$. The first carry is forced and each later no-carry digit has at most $(p+1)/2$ choices. Each permissible residue appears $O(X/p^L)$ times in the parameter interval. Since $(p+1)/(2p) \leq 4/7$, the bad-incidence count per root is at most

$$\frac{AX}{p} \exp\left(-c_0 \frac{\log X}{\log p}\right), \quad c_0 = \log(7/4),$$

for a fixed absolute constant A (210 is fixed). The partial summation in Lemma 6.1 of the supplied draft[1] bounds the normalized total over six roots by a fixed constant times

$$\int_{1/\delta}^{\infty} e^{-c_0 v} \frac{dv}{v} =: \varepsilon(\delta)/A',$$

up to a term tending to zero with X . In particular $\varepsilon(\delta) \rightarrow 0$ as $\delta \downarrow 0$. The argument uses root lifting, not independence among the six factors.

4 Uniform joint carry estimates

For a fixed prime and shift, $ap - c \equiv 195 \pmod{210}$ implies $a = a_0 + 210b$. The variable b runs through an interval of

$$L_{p,c} = X/(210p) + O(1) \quad (10)$$

consecutive integers. Fix $\delta, \eta > 0$, $J \geq 4$, and a nonzero integer frequency vector $h = (h_2, \dots, h_J)$. Suppose

$$X^\delta \leq p \leq X^{1/2-\eta}, \quad p \leq X^{2/J-\eta}. \quad (11)$$

In the phase $\sum_{j=2}^J h_j n/p^j$, remove the integer-valued variation of $h_2 a^2/210$. If all higher frequencies vanish, the phase is linear with slope $h_2 d_c/p$; its normalized geometric sum is $O_h(p^2/X) = o(1)$.

Otherwise, with $j_0 \geq 3$ the smallest nonzero higher frequency, the remaining second derivative is

$$420 \sum_{j=3}^J h_j p^{2-j} = 420 h_{j_0} p^{2-j_0} (1 + O_h(p^{-1})).$$

The first nonzero term dominates for fixed h and growing p . The standard second-derivative estimate[3], divided by $L_{p,c} \asymp X/p$, gives

$$O_h \left(p^{-(j_0-2)/2} + p^{j_0/2}/X \right) = O_h \left(X^{-\delta/2} + X^{-J\eta/2} \right) = o(1).$$

These estimates are uniform in the primes in (11) and all six shifts. The fixed finite trigonometric majorant/minorant argument of draft Lemma 7.1 converts them into uniform box counts. Thus

$$\frac{1}{L_{p,c}} \# \{a : \{n/p^j\} < 1/2 \ (2 \leq j \leq J)\} = 2^{1-J} + o_{\delta,\eta,J}(1). \quad (12)$$

No independence of digit levels at a fixed prime is asserted. The Fourier frequencies are fixed first, so the exact relations with frequencies growing with p do not contradict this argument.

For $X^{1/2+\eta} \leq p \leq X^{2/3-\eta}$, the single phase hn/p^3 has second derivative $420h/p$. Its normalized exponential sum is $O_h(p^{-1/2} + p^{3/2}/X) = o(1)$, uniformly in all six shifts. It follows that

$$\frac{1}{L_{p,c}} \# \{a : \{n/p^3\} < 1/2\} = 1/2 + o_\eta(1) \quad (13)$$

in this intermediate range.

5 The signed residue calculation at large primes

If $p \geq X^{1/2+\eta}$, then $a/p = O(X^{-2\eta})$. In (8), the leading fractional part is $s/210$, where

$$s \equiv (195 + c)^2 p^{-2} \pmod{210}, \quad 0 \leq s < 210.$$

At $s = 0$ the fraction is below one half exactly when $d_c > 0$. At $s = 105$ it is below one half exactly when $d_c < 0$. Away from these boundaries it is below one half exactly when $0 < s < 105$. These signs are uniform for large X because $d_c ap - 210i_c$ has the sign of d_c as $ap = m + c \rightarrow \infty$.

The 48 units modulo 210 have six square classes, each of multiplicity eight. The table lists the numerator s for each of the six shifts; its final column counts the shifts which fail to supply the second carry.

$p^{-2} \bmod 210$	$c = 0$	1	-14	15	-20	21	bad count
1	15	196	1	0	175	36	3
79	135	154	79	0	175	114	1
109	165	154	109	0	175	144	0
121	135	196	121	0	175	156	0
151	165	196	151	0	175	186	0
169	15	154	169	0	175	204	1

In particular the zero column $c = 15$ is *not* bad: its negative correction puts the fraction just below one, not near zero from above. The average bad count over prime unit classes is the exact rational value

$$b_3 = \frac{8(3 + 1 + 0 + 0 + 0 + 1)}{48} = \frac{5}{6}. \quad (14)$$

This table is derived from exact modular arithmetic, not fitted to scan data.

6 Prime summation, boundary coverage and completion

For fixed unit $v \pmod{210}$ and $0 < \alpha < \beta$, the unconditional fixed-modulus reciprocal-prime asymptotic[4, eq. (6)] gives

$$\sum_{\substack{X^\alpha < p \leq X^\beta \\ p \equiv v \pmod{210}}} \frac{1}{p} = \frac{1}{48} \log \frac{\beta}{\alpha} + o(1). \quad (15)$$

This is the classical fixed-modulus formula, not the GRH-dependent varying-modulus estimates later in that reference. An upper endpoint $2X + 21$ is allowed with $\beta = 1$. By (10), the total $O(1)$ progression errors contribute $O(\pi(2X + 21)/X) = o(1)$ after normalization. On primes $p \geq X^\delta$, a uniform conditional error $o(1)$ sums to $o(1)$ because the reciprocal-prime sum is bounded at fixed δ .

Below $X^{1/2}$, omit narrow exponent-boundary neighborhoods and use the bands

$$X^{\max\{\delta, 2/(J+1)\}+\eta} < p < X^{2/J-\eta}, \quad J \geq 4, \quad 2/J > \delta.$$

There are finitely many at fixed δ . A bad valuation-one prime must have no carry at levels $2, \dots, J$, so (12) bounds its cost per shift by $2^{1-J} \log((J+1)/J) + o(1)$. Summing six shifts and then letting the cutoff shrink gives at most

$$6 \sum_{J \geq 4} 2^{1-J} \log \frac{J+1}{J} = \frac{3}{2} S.$$

For the intermediate primes of (13), only the residue-selected bad shifts can obstruct. Equation (15), the table, and (13) bound their cost by

$$\frac{b_3}{2} \log \frac{2/3 - \eta}{1/2 + \eta} + o(1).$$

For $p > X^{2/3-\eta}$, count every incidence in a residue-selected bad shift, without imposing a further carry condition. The cost is at most

$$b_3 \log \frac{1}{2/3 - \eta} + o(1).$$

As $\eta \downarrow 0$ these two costs sum to $b_3(\log(3/2) + \frac{1}{2}\log(4/3)) = (5/12)\log 3$.

For explicit coverage, let

$$\mathcal{E}_\delta = \{\delta, 1/2, 2/3\} \cup \{2/J : J \geq 4, \delta \leq 2/J \leq 1/2\}.$$

For $0 < \eta < \min(\delta/2, 1/12)$ the union of the bands $[X^{\alpha-\eta}, X^{\alpha+\eta}]$, $\alpha \in \mathcal{E}_\delta$, has total normalized incidence cost at most

$$6 \sum_{\alpha \in \mathcal{E}_\delta} \log \frac{\alpha + \eta}{\alpha - \eta} + o(1).$$

This tends to zero as $\eta \downarrow 0$ at fixed δ . The lower bands partition the remaining exponents between δ and $1/2$; the intermediate and large ranges cover the rest. There is no uncovered exponent interval.

Take $X \rightarrow \infty$ first with P, δ, η fixed. The failure limsup is at most the sum of the fixed/repeated cost $O(1/P)$, the small-prime cost $\varepsilon(\delta)$, the boundary cost just displayed, the finite lower-band sum, and the intermediate/large-prime costs. Next take $\eta \downarrow 0$, then $\delta \downarrow 0$, then $P \rightarrow \infty$. The resulting upper bound for failure is $(3/2)S + (5/12)\log 3 = 1 - \rho_3$.

Finally $X = 210T$ gives the t interval $[T - 195/210, 2T - 195/210]$, differing from $[T, 2T]$ by $O(1)$ integers, and $N_X = T + O(1)$. Taking complements proves (4). Strict positivity needs no decimal evaluation: $S \leq 1/4$ and $\log 3 < 11/10$ give

$$\rho_3 \geq \frac{5}{8} - \frac{5}{12}\log 3 > \frac{1}{6} > 0.$$

The polynomial H is strictly increasing for $t \geq 0$, so the successful parameters give infinitely many distinct n . This concludes the proposed proof.

7 Finite checks and remaining review

Two implementations use different factorization and carry computations: Python trial division with explicit digit addition, and a C++ smallest-prime-factor sieve with Legendre sums. They agree on the first 1,000 parameters and on a separate fixed 1,000-parameter high-range comparison. The larger C++ scans give

Parameter interval	valid parameters	observed fraction
$[0, 1000)$	126	0.12600
$[10000, 20000)$	1880	0.18800
$[100000, 200000)$	21456	0.21456

The cross-check on $[199000, 200000)$ agrees in all 1,000 cases, of which 240 are valid. At $t = 61$, $m = 13005$ and $n = 805440$; both methods certify

$$(805443!)^2 \mid 1610880!.$$

The prime-by-prime certificate is saved separately. A literal-factorial check for every $0 \leq n \leq 500$ agrees with the binomial criterion and carry test. These statements are finite arithmetic evidence, not a numerical proof of the Fourier estimates, prime sums, or limit theorem.

The highest-priority independent checks are the six-factor adaptation of root lifting, the signed large-prime table (especially its zero column), fixed-modulus prime weighting, and the uniform-error and limit-order arguments. The next unresolved extension is not obtained by merely incrementing k : one would need to control $n + 4$ as well. This document makes no assertion for $k \geq 4$.

References

- [1] Supplied candidate manuscript, *A quadratic-family approach to the $k = 2$ case of Erdős problem 727*, September 16, 2026, 12 pages. Preserved at `papers/quadratic-family-expanded-2026-09-16.pdf`.
- [2] Companion continuation, *A stronger parameter bound in a different family: a generalized continuation for Erdős problem 727, $k = 2$* , September 17, 2026. Preserved at `papers/generalized-families-2026-09-17.pdf`.
- [3] O. Robert, *On van der Corput's k -th derivative test for exponential sums*, Indagationes Mathematicae 27 (2016), 559–589, Theorem 1. DOI: 10.1016/j.indag.2015.11.009.
- [4] D. Keliher and E. S. Lee, *On the constants in Mertens' theorems for primes in arithmetic progressions*, arXiv:2306.09981v2 (2024), equation (6), page 2. The unconditional fixed-modulus formula only is used.